databricks

Introduction to Continuous Deployment (CD)

LECTURE

Deploying Databricks Assets Overview

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In this lecture, you'll get an overview of deploying Databricks assets, exploring deployment options, Databricks Asset Bundles, and how to integrate them into a development and CI/CD workflow.

Deploying Databricks Assets Overview

Deployment Options

REST API

Databricks CLI

Command-line interface that wraps the REST API
Ideal for one-off tasks, experimentation, and shell scripting

Databricks SDKs

- Available for multiple programming languages (Python, Java, Go, R)
- Allows development of applications, custom Databricks Lakeflow Jobs, and robust error-handling
- Programmatic way to interact with Databricks resources

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Key comparisons:
- Ease of use: SDK > CLI > REST API
- Flexibility: REST API > SDK > CLI

REST API
- Postman and Databricks

CLI
- Command line

SDK
- Python, Go, R, Java

REST API: Most flexible but complex—best for custom integrations.
CLI: Simplifies REST API operations but has limited flexibility.
SDK: Most developer-friendly—best for embedding Databricks functionality in applications.

Databricks Asset Bundles (DABs)

Databricks recommends Databricks Asset Bundles for creating, developing, deploying, and testing jobs and other Databricks resources

Databricks Asset Bundles

Version Control

The practice of tracking and managing changes to code and other development artifacts over time.

Code Review

Systematic examination of source code with the goal of identifying and squashing bugs, improve quality, and enforce coding standards.

Testing

The process of validating expected output from relevant functions and adhering to predetermine requirements.

Continuous Integration

The process of automating development, testing, and deployment to ensure reliability.

Software Engineering Practices

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Databricks Asset Bundles (DABs) are designed to facilitate the adoption of best practices in software engineering, particularly for data and AI projects. Here we will identify 4 core components of SWE practices that are supported by DABs.

Version Control: How are we tracking changes and maintaining a history of code modifications?

Code Review: How are maintaining code quality and are we adhering to coding standards?

Testing: Is our coding behavior consistent and predictable?

Continuous Integration: Are we automating various processes for integrating code changes within our repository?

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Software Engineering Practices

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Databricks Asset Bundles provide a structured approach to managing Databricks projects while adhering to software engineering best practices. By combining infrastructure-as-code principles with automation capabilities, they streamline collaboration, improve quality assurance, and enable efficient delivery of data-driven solutions.

- DABs integrates seamlessly with Git-Based Lakeflow Jobs, enabling users to version their Databricks resources alongside source code
- By treating Databricks resources as code, DABs enables peer review through standard Git Lakeflow Jobs like pull requests
- Developers can use the Databricks CLI with DABs to run tests on bundles in isolated environments, ensuring that Lakeflow Jobs behave as intended
- DABs integrate with CI/CD tools like GitHub Actions or Azure DevOps to automate validation, deployment, and execution of Databricks Lakeflow Jobs

Databricks Asset Bundles (DABs)

What are DABs?

Write code once, deploy everywhere

YAML files that specify the **artifacts**, **resources**, and **configurations** of a Databricks project. This leads to easy configuration of complex notebook and pipeline interactions and **reproducibility** of your Lakeflow Jobs.

What are Databricks Asset Bundles?

DABs are a tool designed to streamline this process for **Databricks projects**. These bundles encapsulate all necessary configurations and artifacts.

How do bundles work?

Bundles provide an exact **definition** of Databricks resources that are to be used within your project with support for **validation** and **deployment** instructions.

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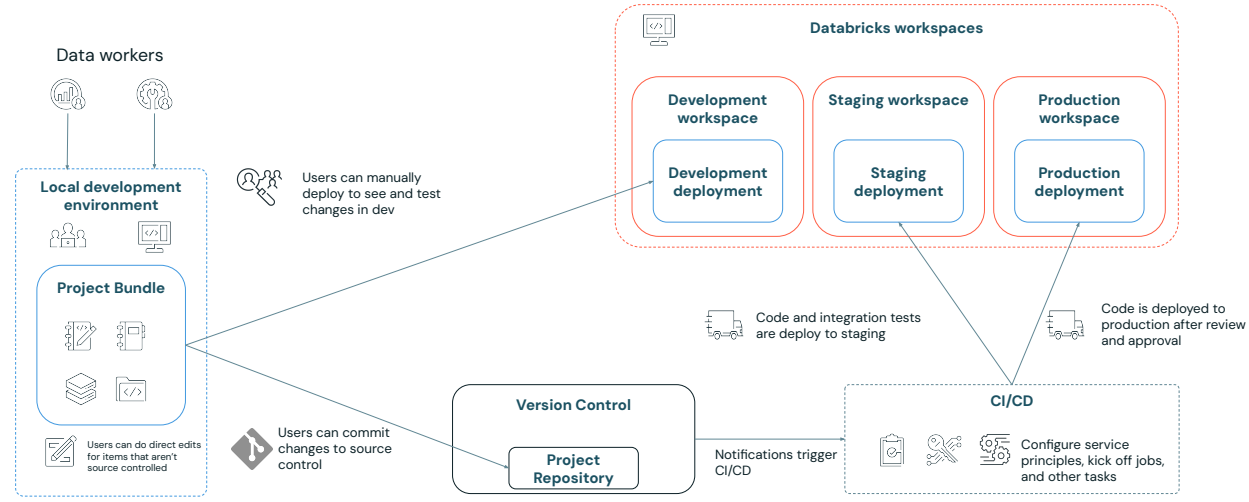
Create code that can be deployed across multiple environments without modification. This ensures consistency, reduces manual errors, and accelerates delivery by automating deployment processes.

DABs are a tool designed to streamline this process for Databricks projects. They enable developers to define Databricks resources (like jobs, pipelines, and notebooks) as source files and metadata in YAML format.

DABs work by first defining your resources and requirements in a databricks.yml file. You then validate the bundle utilizing the Databricks CLI and deploy to your chosen workspace. Once deployed, Lakeflow Jobs or pipelines described in the bundle can be executed.

Development and CI/CD with DABs



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Here we present a high-level view of architecture for development and CI/CD with DABs.

If you're working locally, you build the project bundle with your team using a local environment setup.
Next, you perform version control with project repository, where users commit changes.
Users can manually deploy to test the changes in their development workspace

When users commit changes, a notification is triggered to implement the CI/CD pipeline to staging and production.

In summary, Databricks Asset Bundles (DABs) are a tool designed to simplify the management and deployment of data and AI projects on the Databricks platform. They follow an Infrastructure-as-Code (IaC) approach, allowing users to define and manage Databricks resources—such as jobs, pipelines, notebooks, and machine learning models—through YAML configuration files. These bundles streamline collaboration, testing, deployment, and version control across various environments.

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Thank you for completing this lesson and continuing your journey to develop your skills with us.